

WEEKLY REPORT

1. Project Submission Timeline and Deliverables Status (Phases 1 – 4)

In accordance with the project submission schedule, all engineering tasks and documentation spanning the initial four phases (Weeks 1 to 6) have been systematically completed. Deliverables for Phases 1 through 3 have been committed to the official GitHub repository branch, and the implementation deliverables for Phase 4 (Prototype Development Review) are reported below.

Phase	Duration	Date	Deliverables Required	Execution Status
Phase 1	Week 1	25 Jul – 31 Jul	Problem Identification & Research	Completed (Uploaded to GitHub)
Phase 2	Week 2	01 Aug – 07 Aug	Concept Development & System Design	Completed (Uploaded to GitHub)
Phase 3	Week 3–4	08 Aug – 21 Aug	Hardware/Software Design	Completed (Uploaded to GitHub)
Phase 4	Week 5–6	22 Aug – 04 Sep	Prototype Development (PDR Review)	Completed / Submitted

2. Phase 1 Deliverables: Problem Identification, Objectives & Gap Analysis

2.1 Problem Statement & Context

Reliable communication is essential during emergencies, especially in remote locations such as dense forests, hilly terrains, rural hamlets, and disaster-affected zones. However, these locations frequently suffer from poor or non-existent cellular coverage. During natural disasters such as floods, landslides, earthquakes, and cyclones, commercial infrastructure—including cellular towers, fiber backhaul, and electrical power grids—collapses within hours. This creates an acute communication blackout, severely delaying search-and-rescue operations as trapped victims cannot dispatch SOS alerts or broadcast their coordinates. Existing consumer solutions exhibit severe constraints: mobile phones depend entirely on cellular towers, walkie-talkies have strict line-of-sight range limitations, and satellite emergency messengers require prohibitive hardware costs and recurring subscription plans.

2.2 Project Objectives & Scope

The primary objective of TerraLink is to design and develop a dependable, infrastructure-independent emergency communication system utilizing LoRa mesh networking. Key objectives include: (1) establishing multi-hop LoRa data forwarding across dispersed nodes, (2) acquiring accurate GPS location coordinates during SOS events, (3) maximizing battery longevity via event-driven sleep cycles, and (4) preventing packet broadcast storms through controlled flooding and duplicate packet suppression. The scope encompasses portable handheld user nodes, relay stations, and a rescue base station suited for disaster management, trekking safety, and remote area monitoring.

2.3 Comprehensive Gap Analysis Addressed by TerraLink

- **Dependence on Cellular Towers and GSM Infrastructure:** Standard emergency devices route alerts via GSM/SMS, failing when towers lose power. TerraLink communicates directly over the license-free 433 MHz LoRa band, functioning independently of cellular infrastructure.

- **Dependence on Internet / Cloud Backhaul:** Conventional systems degrade to isolated islands if web backhaul is severed. TerraLink operates locally on embedded firmware with zero dependence on cloud servers or internet gateways.
- **Short Range and Absence of Multi-Hop Relay:** Point-to-point RF devices are blocked by hills or rubble. TerraLink implements decentralized multi-hop relaying, enabling packets to hop across multiple nodes to extend reach beyond direct line of sight.
- **Absence of Integrated GPS Location Sharing:** Many ad-hoc mesh radios lack integrated positioning. TerraLink embeds accurate GPS telemetry (latitude, longitude) directly into the emergency SOS data packet.
- **High Cost and Deployment Complexity:** Satellite terminals and tactical MANETs require complex setups and expensive subscriptions. TerraLink utilizes accessible, low-cost embedded hardware with open-source firmware, making it viable for widespread deployment.
- **Lack of Integrated End-to-End Emergency Platform:** Prior work often addresses isolated elements (only GPS, only RF, or only sensing). TerraLink unifies touch SOS triggers, GPS telemetry, LoRa mesh networking, power management, and local display into a single handheld platform.

3. Phase 2 Deliverables: Concept Development, Brainstorming & Requirements

3.1 Brainstorming & Idea Prioritization

During the conceptualization phase, key system features were scored based on feasibility, operational impact, cost-effectiveness, and implementation simplicity (scale 1 to 5). Features scoring highest—LoRa mesh multi-hop communication (18/20), integrated GPS location tracking (19/20), one-touch SOS sensing (20/20), sleep mode power optimization (18/20), and local OLED status display (18/20)—were selected for prototype development.

3.2 User Persona & Empathy Study

The empathy study examined two primary user profiles: trekkers stranded in network-deprived regions and residents affected by severe natural disasters. A key finding was the 'distress of silence'—the acute psychological anxiety felt after pressing an emergency button without knowing if the signal was received. Consequently, TerraLink was designed with two vital user-centric features: a simplified one-touch SOS trigger for individuals in panic, and bi-directional status feedback to confirm that an alert has reached a relay or base station.

3.3 Functional & Non-Functional Specifications

- **Reliability & Forwarding:** Ensures message delivery through hop-by-hop acknowledgments (ACK), automatic retries (up to 3 attempts), and SRAM duplicate cache filtering.
- **Scalability & Traffic Regulation:** Controlled flooding using unique [Source ID + Sequence Number] cache keys and Time-To-Live (TTL = 5) decrementing to avoid packet looping and channel saturation.
- **Energy-Aware Power Profiling:** Optimized power architecture targeting extended battery life on rechargeable Li-ion cells through event-based GPS activation and sleep cycles.
- **Environmental Operability:** Designed to function reliably across harsh outdoor environmental conditions ranging from -10°C to +60°C.

4. Phase 3 Deliverables: System Architecture, Protocol & Circuit Schematics

4.1 Communication Protocol & Frame Architecture

To optimize radio airtime over 433 MHz LoRa (at SF10, 125 kHz bandwidth), the packet architecture was transitioned from verbose ASCII strings (~78 bytes, 650 ms ToA) to a tightly packed binary structure (`#pragma pack(1)`). By storing coordinates as 32-bit micro-degrees (`int32_t`), total packet size was compressed to 24 bytes, cutting Time-on-Air down to 280 ms—a 57% efficiency gain that minimizes packet collisions.

- **Lightweight Route Discovery:** When no established route exists, the sender broadcasts a Route Request (RREQ). Intermediate relays log the reverse path and forward. The Base Station replies with a Route Reply (RREP) along the reverse route prior to SOS packet transmission.
- **Duplicate Packet Filtering:** Each node maintains a cache of recently processed (Source ID + Sequence Number) pairs. Duplicate frames are dropped immediately.
- **TTL Management:** Packets initialize with TTL = 5. Each relay decrements TTL by 1, discarding expired packets to eliminate infinite routing loops.
- **Hop-by-Hop ACK & Retries:** Relay nodes wait for an acknowledgment from the next hop before completing a transaction, executing up to 3 retransmissions if unacknowledged.
- **Self-Healing Routine:** Upon reaching the retry limit without an ACK, the node marks the current link as degraded, invalidates that routing entry, and discovers an alternative neighbor to reroute the pending packet.

4.2 KiCad Circuit Schematics & Pin Interfacing

Detailed schematic designs were completed in KiCad E.D.A. for all operational hardware units: (1) User Transmitter / Sender Node: Interfacing the microcontroller with an SX1278 LoRa transceiver via Hardware SPI (SCK: D18, MISO: D19, MOSI: D23, NSS: D5), a NEO-6M GPS module via Hardware UART (RX2/TX2), a TTP223 capacitive touch sensor on external interrupt GPIO4, and a 0.96" I2C OLED display (SDA: IO21, SCL: IO22), supported by a TP4056 lithium charging module with DW01A protection and a TPS63001 buck-boost 3.3V regulator; (2) Relay Station Node: Designed with an ATmega328P / ESP8266 controller, Ra-02 LoRa transceiver, and MT3608 boost circuit; and (3) Base Station Receiver Unit: Featuring an ESP32 host, AMS1117-3.3 regulator, 2N2222 transistor driver for a loud rescue siren, I2C LCD interface, and a dedicated 3x3 matrix keypad header for operator inputs.

5. Phase 4 Deliverables: Prototype Development Review (PDR)

Phase 4 marks the completion of the working prototype, embedded software unification, custom PCB layout design, and bi-directional communication enhancements.

5.1 Working Prototype & Upgrade to Dual-Core ESP32

The initial hardware setup used a single-core ESP8266 controller. However, during live trials, serial GPS NMEA parsing and display rendering occasionally blocked the main loop, risking missed LoRa radio reception interrupts. To resolve this, the system was upgraded to the 32-bit dual-core ESP32-WROOM-32E microcontroller. The firmware was partitioned into concurrent tasks running across both cores: Core 0 is dedicated to the time-sensitive LoRa mesh communication stack, handling radio interrupts, packet CRC validation, AODV routing updates, and hop-by-hop ACKs. Core 1 handles peripheral management, including TTP223 touch sensor monitoring, GPS NMEA parsing, and updating the OLED display.

5.2 Printed Circuit Board (PCB) Design

In Phase 4, the circuit was successfully advanced from breadboard prototyping into a custom-designed Printed Circuit Board (PCB) layout developed from the verified KiCad schematics. Key engineering considerations incorporated into the PCB design include: (1) Dedicated power rail decoupling, placing 100nF and 10µF ceramic capacitor pairs adjacent to the ESP32 VDD pins and SX1278 supply pins to handle peak 120 mA LoRa TX current surges; (2) Layout isolation separating the switching regulator section from sensitive GPS and RF signal traces to preserve signal integrity; and (3) On-board integration of the TP4056 lithium battery charger with DW01A protection circuitry to provide safe 5V USB charging and prevent over-discharge below 2.8V.

5.3 Receiver Base Station Enhancement: Operator Keypad Integration

To establish bi-directional communication, a 3x3 matrix keypad was attached to the Base Station. When an SOS alert arrives at the Base Station, rescue personnel can enter preset confirmation codes or operational messages (e.g., confirming rescue dispatch). This acknowledgment is transmitted back through the established reverse mesh path to the originating sender node and displayed on the user's OLED screen, confirming help is on the way.

5.4 Hardware & Software Integration Status

The hardware integration successfully validated concurrent operation across all buses: SPI LoRa communication at 8 MHz, UART GPS NMEA streaming at 9600 baud, and I2C OLED display refreshing at 400 kHz without bus lockups. The software integration successfully consolidated the dual-core task scheduler, routing engine, and bi-directional messaging logic into a unified production firmware build.

6. Phase 4 PDR Submission Checklist & Deliverable Compliance

The following compliance matrix maps the current status against every mandatory item specified for the Phase 4 Prototype Development Review:

PDR Official Deliverable	Technical Implementation Summary	Compliance Status
Working Prototype (Version 1)	Functional standalone hardware nodes assembled with ESP32 dual-core MCU, SX1278 LoRa, NEO-6M GPS, and touch sensor.	Verified & Operational
Embedded Software	Modular C++ firmware featuring dual-core FreeRTOS tasks: Core 0 for LoRa mesh routing and Core 1 for GPS/UI tasks.	Pushed to GitHub
PCB Design Status	Circuit transitioned from breadboard wiring into custom-designed PCB layouts in KiCad with decoupled power rails.	Designed PCB
Mechanical Assembly	Component layout arranged for handheld field operation; secured antenna mounting and battery connections.	Assembled
Hardware Integration	Verified multi-bus concurrency: SPI (LoRa), UART (GPS), and I2C (OLED) operating synchronously without brownouts.	Fully Integrated
Software Integration	Merged packet formation, AODV routing, CRC check, duplicate filter, and reverse-hop ACK into a single binary.	Fully Integrated
GitHub Repository Link	Phases 1–3 and Phase 4 modules pushed to `https://github.com/boomi-2005/TerraLink/tree/development`.	Active (development)
Weekly Progress Report	Comprehensive cumulative documentation covering Weeks 1–6 progress and Phase 4 PDR deliverables.	Submitted Herewith